Farm Class Test Strategy:

java FarmTest

1. Create a Farm object with the default constructor

```
Create a Farm object with the default constructor
Farm name: Unknown
, Farm located in: Unknown
, Number of adult sheep: 0
, Number of lambs: 0
```

2. Create a Farm object with the non-default constructor

   - Positive test: with valid field values

```
Create a Farm object with the non-default constructor
Farm name: testFarm
, Farm located in: Unknown
, Number of adult sheep: 10
, Number of lambs: 5
```

   - Negative test: with invalid farmName

```
FarmTest.java:18: error: incompatible types: int cannot be converted to String
        Farm testFarm = new Farm(10, testState, 10, 5);
                                ^
Note: Some messages have been simplified; recompile with -Xdiags:verbose to get full output
1 error
```

   - Negative test: with invalid farmState

```
FarmTest.java:22: error: cannot find symbol
        testState = new State();
        ^
  symbol:   variable testState
  location: class FarmTest
FarmTest.java:23: error: incompatible types: int cannot be converted to State
        Farm testFarm = new Farm("testFarm", 10, 10, 5);
                                             ^
```

   - Negative test: with invalid numberOfSheep

```
Create a Farm object with the non-default constructor (invalid numberOfSheep)
Exception in thread "main" java.lang.NegativeArraySizeException: -5
        at Farm.<init>(Farm.java:30)
        at FarmTest.main(FarmTest.java:28)
```

- Negative test: with invalid numberOfLambs

```
Create a Farm object with the non-default constructor (invalid numberOfLambs)
Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 5 out of bounds for length 5
        at Farm.<init>(Farm.java:35)
        at FarmTest.main(FarmTest.java:33)
```

3. Test one get method: getFarmName()

```
Test getFarmName()
Farm name: testfarm
```

4. Test one set method: setFarmName()

- Positive test: with valid field values

```
Test setFarmName() with valid parameter values
Farm name: testFarm
```

- Negative test: with invalid field values

```
FarmTest.java:50: error: incompatible types: int cannot be converted to String
        testFarm.setFarmName(-5);
                             ^
```

5. Test display()

```
Test display() (Farm class)
Farm name: testFarm
, Farm located in: Unknown
, Number of adult sheep: 10
, Number of lambs: 5
```